All notes

Monday, April 06, 2026 7:30 PM

Slide 1 – title

My name is Jake, and I looked into the field of actuarial science, and specifically the relevancy of confidence intervals in health insurance.

Slide 2 – research question

We've explored in class the idea of confidence intervals and hypothesis testing, which all follow the general idea that the distribution is normal. However, for insurance companies looking to estimate the expected medical claims per person, the distributions skewed. So, what I'm doing is checking for 95% confidence intervals, is it capturing the true mean of the claim cost 95% of the time like it should? I'll go through different distributions and ideas, as well as assessing the consequences if the coverage fails

Slide 3 – context

To understand why this matters, we'll need to understand what we're talking about and what I'll commonly be referring to. In health insurance, when a person goes to the hospital, doctor's, things like that, the insurance company will be billed. This is called a claim, and they will range from minor visits to major expenses like surgeries and hospital stays.
We then have premiums, which is the amount that the member pays to the insurance company per month. Later, we'll see implications with confidence intervals to estimate premiums, but they need to be set at a rate so the insurance doesn't lose money, or so that the rates are competitive and not too high. The actuary's goal is to estimate the expected cost per member from these claims to accurately set the premium prices

Slide 4 – behaviour

The focus is going to be on claims and how they behave. Typically in the healthcare industry, most people will have small claims, like regular checkups, prescriptions. But every so often, there's going to be significantly larger claims like surgeries, hospital stays, things like that that drive super high costs. So the result is that the claim amounts will be right-skewed with a long tail of what we call the expensive ones to be "catastrophic" events. As you might expect, these heavy costs will significantly impact how the model works. There's a slight issue with typical statistical methods, and it's that most will follow normal assumptions and the symmetry of it, which isn't the case

Slide 5 – normal fails

And so we see that because of the skewness, the normal distribution will likely not perform well for these medical claims, missing the rare but expensive claims. So instead, actuaries use distributions to simulate what they might expect out of claims to be. Because data is typically sparse, actuaries have to simulate things instead of solely relying on data to make their ideas. In this, we'll focus on the Gamma and Lognormal distribution, which are common practices for actuaries to model what claims may be. We can set different parameters for these distributions to change the skew, and measure how confidence intervals perform

Slide 6 – distributions

Here are the distributions we used, and I scaled them all so that it's estimating the true mean to be equal to 1. This is to ensure that we can compare the distributions accurately. Realistically you wouldn't set a true mean, but for the purpose of what we are measuring with confidence interval performance, this is necessary. The first is the normal distribution, which is the control case to verify we set up the mechanics properly. What we'll see is that the methods we use should behave properly when measuring using the normal distribution. Then we go to what actuaries may use, starting with some mild skew, which is the top right graph. You can see the tail extend out, but the "extreme" values aren't very high. A distribution like this would be used for something like measuring pharmacy claims, where prescription drugs will never see anything super expensive. Then we go to what a common commercial or higher-risk groups may look like with our lognormal distributions, the top left and bottom left graphs, with different standard deviation values to change the skewness. These values are commonly used values to estimate these groups, so it's not just random, there's research to back it up. With these, we can see even more of a skew, but also how there's some very large claims, driving the skewness up. And then we move to a mixture, which is a combination of 2 lognormal distributions. 95% of it has a moderate skew to estimate the normal group of people with routine expenses, and the 5% has an increased mean and standard deviation to model the "catastrophic events". You can see in the bottom right graph how there are some

values even pushing 140, even when the true mean is 1. You'll also notice it says the mean is 1.18, which is a result of structural bias as a result of the mixture. We'll explore this later

Slide 7 – CI accuracy

Now we shift to the statistical side. We are pretty much attempting to see if confidence intervals can accurately measure the distributions we see above. Confidence intervals in practice help quantify the risk and uncertainty we see in the expected costs, but if the intervals aren't accurate, like we only see the true mean of 1 in the interval 80% of the time in a 95% confidence interval, we're underestimating the risk and may result in pricing too low or not having enough reserves, which is money to pay out claims. On the flip side, which we may not see with these confidence intervals, but in general if we overestimate the risk and assume claims will be higher than they actually are, we may set prices too high. Insurance industries are very highly regulated, with laws on how much companies are allowed to charge, how much they have to have in reserves, and many others, so accurately modeling the risk is important

Slide 8 – simulation

To answer the idea if confidence intervals work under these realistic conditions, I used a Monte Carlo simulation. We set the true mean to be 1 and used the 5 distributions to estimate claims. I used these 5 different sample sizes, which is how many claims will be generated each simulation. These values represent common sizes of groups in healthcare insurance, like employer group insurance. And I sampled 10000 times per distribution, bootstrapping it 499 times (which is of the samples it generated in the 1 simulation, it will shuffle the values, which can repeat or not include some. This will reduce the variability in each simulation but model more accurately and representative of the true distribution. I calculated coverage, which is how often the confidence interval contains the true mean, as well as the width of the confidence interval, bias of the distribution, and the type 1 error, which is just 1 minus the coverage.

Slide 9 – CI methods

I also used three types of confidence intervals. The first is the z-interval, which always assumes a large enough sample size and a critical value of 1.96 in estimating the interval. I used a t-interval, which instead choses the critical value depending on the sample size. The critical value will be large at smaller samples, but as samples get larger, it will perform like a z-interval. The last is a bootstrap interval, which uses the 2.5^{th} and 97.5^{th} percentile of the values of the 499 bootstrapped resamples to create the interval. Unlike the others, this does not assume normality and won't be symmetric. We can use these intervals to see if any are more accurate in the coverage after running the simulation

Slide 10 – coverage

Here are the 3 graphs of each confidence interval method to see the coverage, how likely it contains the true mean. From all 3, you can see most of the distributions get closer and closer to the 95% confidence interval coverage. The normal distribution, as it should, was almost always the 95%. Gamma was slightly less, but converged to 95% as the sample size increased, Both lognormal distributions were farther off, but got closer and closer as the sample size increased. The reason we used the 3 confidence interval methods was to see how they are different. Almost always, t-interval had the best coverage, followed by z-interval, and worst was bootstrapped. This happens because of the width of the confidence intervals, where the t interval will be wider as the critical value changes from sample size. Bootstrap is the worst because it doesn't follow the normal assumptions, and so the interval width is even right-skewed and misses the true mean, commonly less than the lower bound of the interval. But, each behave similarly, so we won't need to see the difference between the 3 of them. For these distributions, they are all unbiased estimators, so the central limit theorem tells as that for unbiased estimators, as n grows, the distributions will converge to a normal approximation. So essentially it just depends on the sample size if we should use confidence intervals for those distributions. The most interesting is the mixture distribution, which was the only one that decreased coverage. This is because of the structural bias.

Slide 11 – bias

In the simulation results, the mixture almost always overestimated the sample mean by around .25, no matter the sample size. This is because the catastrophic claims pull the mean upward. Now the bias occurs because of the 5% likelihood of the catastrophic claims. For example, if there's only a sample size of 25 and 5% are catastrophic claims, it's expected to be less than 2 of them. But since it's a simulation, it may happen that there could be 3 or 4, so naturally that will increase the sample mean significantly. Even over so many simulations, this happening will pull the sample mean over the true mean and create this bias.

Slide 12 – type 1

We'll shift quickly to the type 1 error, which is just 1 minus the coverage. This is the likelihood of rejecting the null hypothesis, or what we set as the true mean of 1. Now, this isn't a traditional hypothesis test, it's sort of like how often will the hypothesis test falsely reject. It's supposed to be 5% for a 95% confidence interval, but as coverage decreases, we can see the Type 1 error increase. We can make the connection that confidence interval failure and hypothesis testing are essentially the same tests. I want you to notice one thing on the graph, which is how the mixture distribution starts off okay, and gets worse and worse.

Slide 13 – Width

This is because of the confidence interval width. Now we can see that lognormal and mixture have very similar widths, which are significantly wider than the others because the skewness increases the variability of the distributions, so why does lognormal have such low coverage on such a high width? This is because only sometimes, the sample mean is way overestimated on the catastrophic events, and the width can't compensate for it. Now you may ask why that's not the case for the mixture. Well, the mixture is more consistently at its sample mean, and it's almost as if the interval accidentally captures the true mean of 1. Because of the bias, as the number of samples increases, the confidence interval becomes more narrow and centers more around the sample mean, so will miss the true mean a lot more often. For the other unbiased distributions, the confidence intervals get more narrow but centered around the true mean, and the increased sample size reduces the variance, which is why these unbiased estimators will eventually converge to the 95% coverage

Slide 14 – what it means

What does these results actually mean for actuaries. Are we able to use standard confidence intervals to estimate claims? Well, sometimes. We saw for the gamma distribution, at small samples like 25, it was close but not enough, but as the sample size increase, it did work. As we increased the variability to the lognormal distributions and the mixture, we saw how even at large sample sizes of 500, it never fully had 95% coverage, so we'd always be underestimating the claim amounts of we used confidence intervals. For our most realistic mixture distribution, confidence intervals completely fall apart and wouldn't be reliable at all. We can see that in general, these methods should not be used in practice

Slide 15 – implications

As we said before, we don't want to underestimate risk with the lower coverage, as we can end up with pricing the premiums too low and have higher claim costs. A lot of these results depend on the sample size, but in practice there could be all sorts of different sized groups, so using confidence intervals will always depend on what the distribution of claims may look like and how large the group is. In an actuarial setting, we should consider alternative methods for estimating claim amounts to more accurately measure costs and risk.

Title: Do normal confidence intervals that we know, even from this class, accurately show the expected medical claims per person?

We're looking into the healthcare industry, specifically into costs for insurance companies when they have to estimate costs of covering patients.

The motivation behind this project is I am pursuing an actuarial career, and in the field of it on the health side, an actuaries goal is to essentially predict the expected costs of people needing this medical attention so that the healthcare company or insurance company doesn't lose money can actually function. If we underestimate the expected cost pet per member, the company can lose money, but if we overestimate, then the amount charged, called a premium that a person pays monthly to an insurance company is too high and may lose customers. That's a very baseline definition, but for the purpose of this project, the rest of it isn't too important to know, but if you are interested, you can see on your papers a few quick notes on what an actuary does in the healthcare industry and you can do research from there.

Before we get started, we have to understand the general idea of how it works in the medical industry. When a person goes to the hospital or whatever it is, the bill they get is sent to the person's insurer. Most people have health insurance, and so the bill submitted is called a claim. These claims always very depending on the service, so things like a doctor's visit can be relatively cheap, like $150, or you can get something more expensive like a surgery. The difference in costs per claim is what's known as the claim severity, and actuaries have to estimate using the severity to find what the total cost per member is. In general, what happens is that most people have very small costs, but there are a few very large costs.

What arises with this severity is that costs don’t follow a normal distribution, and in fact, there's really no perfect distribution to estimate this. These medical claims are right-skewed with long tails and include outliers, which we call catastrophic events. I'm not going to go into too much on what the distributions are, but I've provided a brief summary for you guys with these 4 distributions – Gamma, Lognormal, Weibull, and Pareto, which are common practices to use in the actuarial field. These all help

model outliers and high-cost claims where distributions like the normal distribution may not cover, and so these types or even a mixture of them help define the distribution of claims.

There's a couple things in statistics that in practice for insurance companies and with these medical costs may or may not work. Obviously first one we talked about with the distribution being normal doesn't work, and so using the Central Limit Theorem and "Large enough" sample sizes may not be feasible. So we can explore standard confidence intervals with these ideas. For actuaries, confidence intervals help quantify the pricing risk. Since the expected costs change over time each time period, actuaries need to make sure a company is prepared for the worst, and they take risks into what is plausible through this range of possible expected claim costs. The goal of this is the find if confidence intervals are still realistic to use these methods in the actuarial practice, or what we need to do to adjust so that we can more accurately use confidence intervals. Because of small sample sizes for small insurers, extreme values, and the skewness, we can explore if and how confidence intervals break when we model claims.

You may be asking what the implications are if you use some method like a confidence interval isn't accurate. In an actuarial practice especially for insurance companies, risk management is the idea, and actuaries have to determine the acceptable amount risk for the company to take, like how much reserves to keep to be able to pay out claims, thigs like that. If we determine a 95% confidence interval but it only covers like 80%, we underestimate the risk and could cause implications, like legal ones of not having enough money to pay out claims, or profitability standards that could result in legal trouble. In a heavily regulated thing like insurance, measuring the risk is very important.

To shift to the project, we ask the question: do standard confidence intervals actually achieve their intended coverage when applied to realistic medical claim distributions?

RESULTS

Statistics calculated on the results:
Distribution – the mean for all distributions should be 1 to accurately compare them – combine with [background knowledge] section about distributions

- Health claims never follow a specific distribution, because there's always changing factors such as technologies, patient types, inflation, and more. But there is actuarial literature that helps show what common shapes of the distributions there and what parameters are reasonable. We're not calibrating to a specific dataset, but rather choosing parameters that would be reasonable in the real-world to select, and we can compare what happens under different models that show different severities
- Normal (mean=1, variance=1) - the normal distribution is chosen as a benchmark distribution to essentially show that the 95% confidence interval follows normal assumptions. There's no realistic health data that will follow this distribution, it only serves as verification of the CI
    - It is reasonable to assume every interval will be close to .95 coverage. At n=25, we see slight deviation from it, like z-interval being .94, which may have to do with the critical value being 1.96 as it assumes the number of samples is large enough, which 25 may not be. In fact, we'll see for every distribution that the t-interval will almost always have more coverage than the z-interval, especially at lower samples, for the "large enough" sample size reason. Bootstrap also varies at .931 because the small sample size creates some variation, and the bootstrap interval follows only a rough approximation of the true Normal (at larger values, it will follow a normal distribution and behave similarly to confidence intervals, which follow normal assumptions)
    - As it approaches larger sample sizes, it gets closer and closer to .95 coverage for all methods, which we should expect
- Gamma (mean=1, shape=2) - a shape of 2 is essentially setting the skewness and variability of how extreme the values might be. In practice, the Coefficient of Variation CV=1/root(shape) = .71, and its common to have around this value to model the realistic range of some group of claims. It's skewness is calculated to be 1.41. This distribution is commonly used for claim severities like pharmacy claims, which tend to have a moderate skew and not as large values.
    - This distribution is where some realistic health claims may follow. At smaller sample sizes like n=25, we see t=.934, z=.921, and bootstrap=.912, which is a little farther away than the normal distribution. As we increase the sample size and get to n=500, all 3 methods are roughly .95, so we can conclude that while maybe not true for small samples, if there's a large group of like pharmacy claims, it can be reliably calculated using classical methods
- Lognormal (mean=1, log-str.dev = .75) - In practice, the lognormal distribution is one of the most used and studied in insurance claim modeling. The log-standard deviation of .75 is in the middle of the range for typical commercial health population's claim severity. It creates a larger right tail, with its skewness being 2.5 (notice its greater than gamma distribution, so it will have more extreme values)
    - This is the distribution where you'll find heavier of a tail and some more extreme values. At n=25, we see the 3 methods all around .90 coverage, and as we increase, to n=500, it only reaches about .945. Here, we slightly see that classical methods start to fail. While not considerably off, it does tell us we would be underestimating the risk slightly if we try to use

classical methods, so it shouldn't be used here. Keep in mind that this is the distribution where we start to use it for general commercial insurance, which will be a key piece in our conclusions at the end

- Lognormal (mean=1, log-str.dev = 1.5) - This is a more aggressive choice which typically lines up in the range of commonly used log-standard deviation values for high-cost claimants. An example is that this distribution might be used for a a group of people that are considered higher risk, so like older individuals or ones with underlying health issues. Skewness jumps to 11
    - Now, we start to see a huge shift in our coverage, where our 95% confidence level is actually only about 77% of the interval for all 3 methods at a sample size of 25. Even as we increase the sample size to 500, we only get about 90% coverage, and we can really start to see how the heavy skewness for claims is shifting our confidence intervals away from the true mean of 1
- Mixture (95% Lognormal(mean=.8, o = .75) + 5% Lognormal(mean = 4, o = 2) - This distribution is the most justifiable, where a popular research group called the Kaiser Family Foundation and other sources have documented that the 95/5 split is the most consistently used split. It is said that approximately 50% of costs from claims come from the top 5%, which is noted as the "catastrophic" group (people with major surgeries, cancer, long-term hospitalizations). It is deliberate that a high log-standard deviation is used for this 5% because it is meant to model that worst case conditions that may be encountered, and the 95% has a lower mean because it represents the majority of the population that will only likely have routine claims
    - The mixture is interesting and very different from the other distributions, and almost appears backwards. At lower sample sizes like n=25 and n=50, the coverage appears okay. But there's one thing very different about this, and it's when you look at the bias, which is .28 at n=25 and .22 and n=50, which essentially means the sample mean is underestimating the true mean by 22-28%. You may be asking, why is the coverage so high then? We can shift our attention to the average width of the CI, and we can see that at n=25, it's about 1.55 width compared to a normal distribution's 0.85 width. So basically, the interval is almost just accidentally capturing the true mean just because of how wide it is.
    - To understand why these things happen, we have to understand how the mixture works. Say we are doing a sample size of 25. With only 5% catastrophic claims, it's expected that 1.25 claims will be large. But, sometimes it may be 0, sometimes may be 2, 4, even 5. So when it happens that there's more than a couple is that it pulls the mean higher than it should, way above the true mean of 1. So, these intervals where there are many catastrophic claims shift the bias upwards, and so now it's like the sample mean is overestimating the true mean by around .25 for all sample sizes. As the sample sizes grow, the bias stays the same, but the interval shrinks because of how the variability gets smaller Now we're seeing that at larger sample sizes, we're centered on average around something that's not the true mean, and the confidence interval is narrow where it won't capture the true mean accidentally. We're seeing coverages significantly drop all the way to around .6 for t and z-interval, and .36 for bootstrap, as the sample size increases and confidence interval gets smaller
    - You might ask after seeing the width of the second lognormal distribution that it's nearly identical to the mixture, so why is that so much lower coverage in the beginning but increases later? It's because with small sample sizes, the sample means jump around a lot because of the skewness and high variability. So often times, around 77% that we found before at small samples, it's missing, but as large sample sizes are taken it starts to center itself around the true mean more and more
    - The mixture is unique because of its bias. Even though it has high variability, the idea of the bias overestimating the mean and essentially centering itself around a higher value shows why it decreases coverage rather than increases.

N Sample sizes – following real actuarial context

- A common plan that insurance companies cover is health insurance in the workplace, like an employer offering health insurance to their employees. In fact, over 50% of people with health insurance do it through their employer. So, actuaries often model per group or per company. Another reason for grouping would be modeling a group of people in certain "risk" classes. When people file for health insurance, they are grouped in 1 of 4 classes depending on their health conditions and other factors like age, location, etc.
- N=25 and n=50 are what's known as the small groups, where groups of this size are often much more volatile because of the small sample sizes. We know this in practice that simply having more people result in smaller variance. You'll see with these smaller sizes that typically, it's less likely for the confidence interval to contain the true mean of 1
- As the sample size increases to 100, 250, and 500, are common size points that are modeled at. Here, and as we grow larger to 500 or even more, it is assumed that asympototic results are reasonable and that it converges it it's "coverage" value

Methods – the distribution methods to determine the coverage if the true mean is inside the interval

- Z-interval – always assumes 1.96 for a 95% CI, whereas t-interval is dependent on the same size. They are very similar, but for smaller sample sizes the CI will be wider for the t-interval
    - Formula is "Mean +- (Crit.Value) * (str.dev / root(n))

- Bootstrap doesn't use any formula, it just takes the boundaries. So it will find the 2.5 percentile and the 97.5 percentile of the means as the confidence interval. It doesn't assume any shape or distribution
- For each simulation (1 of the 10000 per sample size per distribution), the code bootstraps it 499 times. Basically it just mixes and randomly chooses n samples based on the sample size from the number it generates, and it can repeat or leave out some generated claims each bootstrap. Then it calculates the mean and stuff (see code on right)

Coverage – Of the 10000 simulated samples for each distribution, what proportion of the confidence intervals actually contained the true mean. Since we did 95% Confidence Intervals, we would expect 9500 of the intervals to capture the true mean, but if I drops below that number, then the method is overconfident. In an actuarial perspective, if it's like 85% coverage but we say 95% sure with that confidence level, then we are pricing with less certainty than we actually have and its essentially just underestimating the risk

Average Interval Width – shows how wide the 95% confidence intervals are, simply with the upper bound – lower bound. If an interval is wider, it may have more coverage but it's less useful, because it varies a lot, whereas if the interval is narrower, it's more precise but risks missing the true mean

Bias – The (sample mean – true mean), which tells is if the same mean consistently over or underestimates the true mean. For a sample mean with set, full distributions, you'd expect it to be an unbiased estimator, which is true. As only one distribution, it will always have the mean accurately. But when we get to the Mixture, the bias values are not 0. Why is that?

Mean Squared Error (MSE) - = Bias^2 + variance. This answers how far off the estimate is from the true mean. Notice the common trend that MSE shrinks as n grows. Why does this change for Mixtures?

Type 1 error – Simply (1 – Coverage). This allows us to use hypothesis testing, by setting the null hypothesis to the true mean.

If we look into the scope of the project, I'm going to be doing a simulation for this data. The reason for this is that with a real dataset, we can't evaluate the coverage probability, which is just the probability that the insurance company will be able to cover and pay out all the claims. We need a true mean, and if we used a dataset, we'd only be given one sample of data from one source. So a simulation allows us to repeat sampling many times, change the number of data points per sample (can be used to simulate a smaller company that receives less claims), and we can also measure the variability and bias. It will help model realistic claim behavior which we may not get with one dataset, unless large enough, but datasets like that aren't exactly publicly available.

We'll be simulating using Monte Carlo methods, which will generate these sample claims based on the factors we set, such as the distribution, number of samples, and more, which we'll go through during the modeling process. From the generated data, we'll calculate point estimators, confidence intervals, and use hypothesis testing to analyze results, which will be dependent on the different parameters we choose for the simulation

Understand for the distributions

- Mean – some target parameter that we set the same for every distribution. Project revolves around estimating this
- Variance – Controlling spread and affects CI width
- Skewness – The asymmetry of the distribution, like long-right tails (what breaks the normal confidence intervals)
- Tail behavior – The likelihood of extreme values (impacts small-sample inference)

Distributions used

- Normal – Control case
  - We are testing classical confidence intervals, which utilize the normal distribution. It's what we explored in class, but in the actuarial profession, is this reasonable?
  - Do we get 95% coverage, meaning does the 95% CI actually cover 95% of data
  - If code is done right, CIs work for normal distribution
- Gamma distribution – Moderate skew
  - A commonly used distribution for positive, skewed heath data
  - Realistic, but not extreme
  - Does some mild skew already break inference
- Lognormal – Moderate variance and high skew – 2 cases
  - Used because healthcare costs grow multiplicatively

        - Means that factors like inflation or other costs cause a percent increase/decrease for a service, not a specified additive amount
        - Log transformation is normally distributed, so allows simplified analysis while preserving typical skewed health claims
    - Tests how bad under coverage gets as variance increases
- Mixture – 95% normal, 5% catastrophic
    - Insurance portfolios often behave like this, where most are chap and a few dominate costs
    - Creates extreme skew and variance, instability in sample mean
    - Tests worst-case scenario

Confidence Intervals – Core statistical insight

- How skewed can the data be before these assumptions fail

Bootstrapping – resamples the claims however many times. It can repeat observations, skip some, whatever of the number of claims.

- Because the data for most of the distributions are not normal, bootstrapping is used to reflect uncertainty. Sample means may vary among bootstrapped data. From the different same means, it creates a confidence interval based on the bootstrapped values.
- The formula for the CI is that it assumes the data is bell shaped, but the bootstrap will see some values and pull the mean. After it creates the confidence interval, it sees if the mean, which is 1, is actually in it

Calculates

- Z-interval – always assumes 1.96 for a 95% CI, whereas t-interval is dependent on the same size. They are very similar, but for smaller sample sizes the CI will be widerollow normal approximation
    - Formula is "Mean +- (Crit.Value) * (str.dev / root(n))
- Bootstrap doesn't use any formula, it just takes the boundaries. So it will find the 2.5 percentile and the 97.5 percentile of the means as the confidence interval. It doesn't assume any shape or distribution
- For each simulation (1 of the 10000 per sample size per distribution), the code bootstraps it 499 times. Basically it just mixes and randomly chooses n samples based on the sample size from the number it generates, and it can repeat or leave out some generated claims each bootstrap. Then it calculates the mean and stuff (see code on right)

The reason these different types of distributions were used is because each represent a different mechanism that generates skewness and tail risk, and affects inference

- Gamma behaves more additively, as it sums exponential variables
    - We know the exponential distribution and the Poisson distribution. Gamma combines them
    - Reason it's used is because it measures more of the accumulating small costs, not necessarily the compounding effects of what drives cost
- Lognormal factors the multiplicative risk structure
    - As there's complications, additional procedures, longer stay, etc. These costs grow almost exponentially rather than additively.
    - Results in a much heavier tail with extreme values far more likely, sample mean influenced by rare events
- If we look at confidence intervals between these two distributions, we're more looking if the inference methods fail when the skewness comes from compounding vs accumulation
- The Mixture distribution is conceptually different as it combines subpopulations, which we define as the regular claims vs catastrophic claims
    - Extreme values come from a different process rather than a set process in a distribution
    - Can interpret as like 95% are normal routine healthcare, like visits and minor things, and the 5% come from like major surgeries, cancer stays, etc.
        - Both A and B are its own distributions
            - An example is that if it comes from the 95%, it draws moderate lognormal, but if it's in the 5%, it draws from extreme lognormal
            - These are the "different regimes", just more than one underlying distribution it comes from
        - Mixtures will create huge variance – in a sample, especially smaller ones, sample means will change a lot because there may be no catastrophic claims or there may be multiple, so it's even more random that the other distributions that come from a single process
            - True mean is still what it's set at (like 1), but E[X] = .95*E[X1] + .05*E[X2], so mean for X1 can equal .8, but mean for X2 can equal 5, whereas the other distributions all just had mean 1
    - This reflects real insurance risks, especially among smaller companies, when small number of claims can lead to huge variability
- We use these different distributions because our goal isn't to actually model, we're evaluating inference performance. Essentially, we're using variation in conditions to see if our statistical method in confidence intervals actually work

Statistical content covered

Distributions and random variables

- The fundamental piece of the project, where we use distributions to estimate an individuals health claim cost. Each distribution measures potential severities.
- This is a controlled experiment where we see the severity variability increase systematically, and measuring of confidence intervals performance degrade accordingly
    - This monotone relationship is a key finding – as variability increases, confidence intervals perform worse

Estimation – sample means, Bias, MSE, Consistency

- Sample mean is the estimator throughout
- Bias – the normal distributions just verify that the bias is 0 which is a theoretical gaurentee, but the Mixture breaks this pattern due to structural sampling problem with catastrophic events
- MSE – for the unbiased estimators, MSE is the variance, and it explains why distributions with high variance in the sample mean from the high skewness and forces intervals to be wide
- Consistency – the estimator converges to the true parameter. This means that as the sample size grows, the confidence intervals alwaus shrink, and follows accordingly with the MSE shrinking

Confidence Intervals – in other spots

Asymptotic Theory and CLT Convergence

- CLT says that regardless of the underlying distribution, the sampling distribution of X approaches the normal distribution, which is a theoretical foundation using z and t intervals
- This project is like a stress test for the rate of convergence, or how fast it will reach the 95% coverage. This is for unbiased estimators, as we see for the Mixture that CLT is irrelevant because it's a structural boas problem, and not asymptotic (that it will converge to the 95%)
- The project is shows that for most underlying distributions, we know it will converge, but how long is the question, because in actuarial studies, the sample size is important and matters which is why we tested different options. Shows us that we can't always use confidence intervals for estimation, but if sample size is large enough or if there's unbiased estimators, then we can

Hypothesis testing

- This is used as a diagnostic lens rather than a decision making tool, because we constructed the null hypothesis of mean 1 to always be true
- Instead, we are looking at the duality of confidence intervals and hypothesis testing. If a 95% CI fails to contain the true mean of 1, it’s a mathematically equivalent procedure to where a two-sided t-test at alpha=.05 rejects the true mean 1. One gives an interval and the other is a yes/no, but derived from the same critical value.
- So we can say the coverage findings translate to hypothesis testing. We can say for example, when LognormalHigh at n=25 has coverage .777, we are basically saying that a t-test applied to 25 high-skew claims following the same distribution would produce a 22.3% false rejection rate instead of the intended 5%. The Type 1 error shows this to make the connection to the duality of CIs and hypothesis testing

Slides:

slide 1: title (Actuarial Statistics - Confidence Intervals in Health Insurance) by Jake, for Math 476

Slide 2: research question

Slide 3: Context about actuarial science and the health industry (that insurance companies cover costs for patients, explaing claims, premiums, and implications of finding expected cost per member

Slide 4: general trends/ideas on why how medical claims work and why they are typically skewed

Slide 5: transitions to why normal doesn't work, and why we have these other distributions such as gamma, lognormal, weibull, pareto. We focus on the two most researched ones, gamma and lognormal. Explain basics of ordering of skewness.

Slide 6: Show distribution of the 5 claims (i have the png), what situations they may be used for. We saw normal doesn't work, gamma for mild skew claims, like pharmacy claims. Lognormal has two ideas, one with moderate skew and one with high skew, to see difference. The last is a mixture, with expected value equal to 1, and uses 95% moderate lognormal with .8 mean, and 5% high skew (higher variance) lognormal with 4 mean. Insurance portfolios often like this as most people are cheap with little things, while a couple have very expensive like cancer stays, high level surgeries, etc.

Slide 7: Shift to our statistics ideas. if we use confidence intervals, what are the implications if the interval isn't accurate in an actuarial context. So then, we ask if these standard CIs achieve intended coverage in realistic medical claim distributions

Slide 8: We set up a controlled monte carlo simulation, setting different sample sizes as it will change how confidence intervals react. We'll measure coverage, which is how often the true mean we set at 1 for each distribution falls in the 95% CI, and we'll perform 3 sampling methods, t and z intervals and bootstrap. Explain each of these. We'll also see variability with the width of the confidence interval, and

measure the bias to show how particularly the mixture is the only one that's not an unbiased estimator. We'll end up explaining how the distributions react under different situations and samples, and if CIs will work.

Slide 9: Maybe can split slide 8 into 2, there's a lot of content, especially if we have to explain t and z intervals and bootstrap methods. Can also put code or show how it was done, or at least explain like 10000 simulations, etc.

Slide 10: look at coverage, our main result. Coverage means that in the supposed 95% CI we are measuring, what proportion of CIs actually contain true mean. Connect back to idea earlier that if it's less, we underestimate risk. Put in coverage chart, explain how there's different sample sizes and that it increases and converges to 95% as sample size increases. From CLT convergence and asymptotic theory idea. Show mixture is biased and it overestimates the true mean by about .25, which is why mixture behaves the way it does. The CI is wide enough early with a low sample size to "accidentally" cover the true mean, but as the interval narrows it covers less, which gets this shape

Slide 11: show bias. Should be quick slide, just that other distributions (singular ones, not mixtures) are a theortical guarantee to be unbiased, but that mixture is different. Some likely end of what i said in slide 10 will go here.

Slide 11: show type 1 error and its reciprocated 1-coverage idea. should be a quick slide, explaining type 1 just means that we miss the interval _% of the time. Connect to hypothesis testing. A 95% CI measures an interval of the true mean but is the same as testing the type 1 error, rejecting the true mean of 1 _% of the time instead of intended 5% (can use example i put in my notes)

Slide 12: reinforce confidence interval width with the plot i have, and say again how it connects to mixture. Also, explain how skewness makes CI wider

Slide 13: explain what these results actually mean. Put up coverage png again and explain how like gamma at very samll sample sizes may not be able to use CIs to capture mean, but at larger ones it is used. As it skewness increses, it becomes less and less reliable. Mixture, one of the more realistic ones in general, show that using CIs severly underestimate risk and would not be reliable in most cases. At small sample sizes, it may appear okay but it's just a disguise as the width doesn't tell us much

Slide 14: Actuarial implications, reiteration on risk and how it impacts pricing. This should be a conclusion slide.